

Impedance Matching Considerations for Ferrite Faraday Rotators

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Abstract—When a linearly polarized plane wave propagates from a dielectric medium into a medium with Faraday rotation, a reflected wave will always occur at the interface. This reflection is necessary because the normal-mode wave impedances in the medium with Faraday rotation are nondegenerate, while the equivalent normal-mode wave impedances for the dielectric medium are degenerate. Therefore it is not possible to match both normal modes simultaneously at the interface. This paper investigates the peculiar reflected wave associated with impedance matching of Faraday rotators, and shows that it should have minimal impact on practical devices.

Index Terms—Faraday rotation, ferrites, impedance matching, microwave propagation.

I. INTRODUCTION

THE FIRST microwave ferrite components, marketed some 50 odd years ago, were simple waveguide isolators using 45° Faraday rotators with fixed longitudinal-field permanent magnet bias. Subsequent development of resonance isolators, junction circulators, axial toroid phase shifters, and other ferrite devices based on transverse-field magnetic bias largely replaced the simple Faraday rotator as a useful microwave component. One exception to this comment is the latching dual-mode reciprocal ferrite phase shifter [1], [2], which uses a longitudinal bias field, but operates with circular polarization in the biased ferrite region instead of the linear polarization used in a true Faraday rotator.

Thus, a close relationship exists between the design considerations for dual-mode phase shifters and Faraday rotators; the same analytical methods can be used for both with differentiation applied only in interpretation of the results. The similarities extend to design details, and up-to-date dual-mode phase shifter geometries have recently been applied [3] to Faraday rotators for use in unique reciprocal microwave ferrite switches.

Although an overview of the basic principle of Faraday rotation is regularly presented in introductory courses on microwaves, the focus is properly on demonstrating the important transmission characteristics of the interaction. A quick search of early literature by this author failed to find any references at all dealing with the impedance matching at the junction of an ordinary medium and a medium providing Faraday rotation. The problem can be presented in elementary terms using simple transmission-line parameters. Carrying out such an analysis

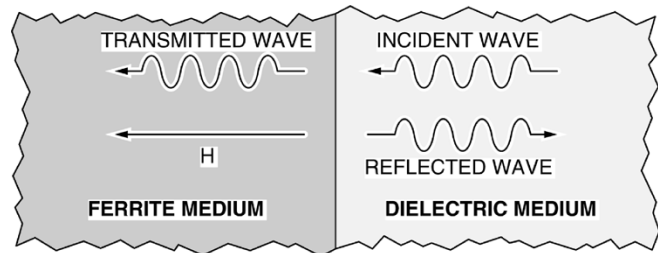


Fig. 1. Plane-wave scattering arrangement.

and examining the consequences for practical Faraday rotation circuits is the objective of this paper.

II. BASIC CONSIDERATIONS

Consider the scattering conditions sketched in Fig. 1. In this arrangement, a linearly polarized plane wave propagating in free space filled with a homogeneous isotropic medium abruptly enters a medium, at normal incidence, in which Faraday rotation occurs. The normal modes of propagation in the Faraday rotating medium are modes with right-hand circular polarization (RHCP) and left-hand circular polarization (LHCP); recall that linear polarization can be expressed as a combination of equal amplitudes of RHCP and LHCP. The space filled with homogeneous isotropic medium will have degenerate RHCP and LHCP modes with the same wave impedance, while the space with Faraday rotation will have nondegenerate RHCP and LHCP modes with different wave impedances.

Now define the relationship between the three values of wave impedance as

$$Z_{0R} = r\alpha Z_0 \quad (1)$$

$$Z_{0L} = \left(\frac{r}{\alpha}\right) Z_0 \quad (2)$$

where Z_{0R} and Z_{0L} are, respectively, the wave impedances of the RHCP and LHCP normal modes in the Faraday rotating medium, and Z_0 is the wave impedance of the degenerate modes in the dielectric medium. The normalization has been chosen such that

$$Z_{0R}Z_{0L} = (rZ_0)^2 \quad (3)$$

and

$$\left(\frac{Z_{0R}}{Z_{0L}}\right) = \alpha^2. \quad (4)$$

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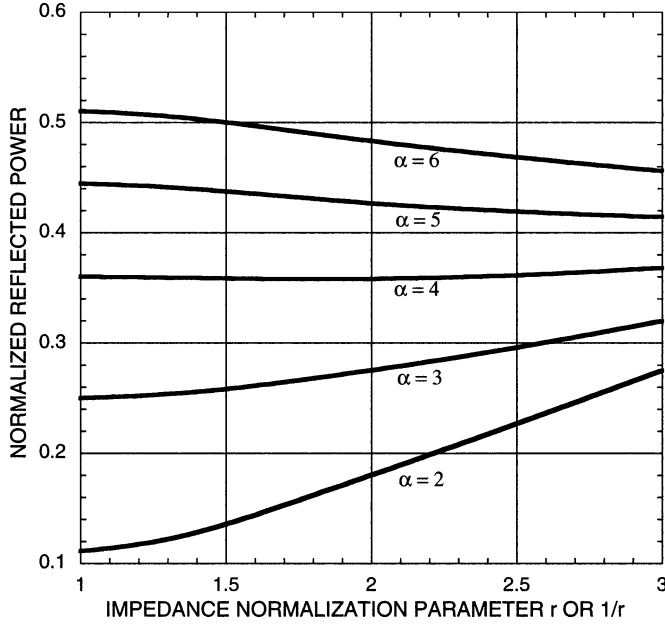


Fig. 2. Reflected power versus $r = 1$ for selected values of α .

The reflection coefficients ρ_R and ρ_L for the RHCP and LHCP waves will be

$$\rho_R = \frac{r - \frac{1}{\alpha}}{r + \frac{1}{\alpha}} \quad (5)$$

and

$$\rho_L = \frac{r - \alpha}{r + \alpha}. \quad (6)$$

Clearly, when $r = 1$, i.e., when Z_0 equals the geometric mean of Z_{0R} and Z_{0L} ,

$$\rho_R = \frac{\alpha - 1}{\alpha + 1} = -\rho_L. \quad (7)$$

The significance of this result is that only for the case $r = 1$, a linearly polarized wave incident on the Faraday rotating medium will produce reflected waves of equal, but opposite amplitude, i.e., the reflected wave field vectors in the plane of the incident wave will cancel. However, a quarter-cycle later, the RHCP and LHCP reflection vectors will add, producing a linearly polarized reflected wave. This peculiar reflected wave will thus be in time quadrature, as well as at a right angle in space, with respect to the incident wave.

For the expected case in which α is not too large, i.e., when the RHCP and LHCP characteristic impedances are not too different, setting $r = 1$ appears to provide a minimum of total reflected power in the two orthogonal linear polarizations, as shown in Fig. 2.

Note that for large values of α , setting $r = 1$ can cause the normalized reflected power to be greater than 0.5. In this case, the normalized reflected power will be reduced to 0.5 or less by exactly matching one of the CP modes and allowing unconstrained reflection of the other CP mode. This case is not likely to occur in practice because the separation between the Z_0 values for RHCP and LHCP is much greater than that which

can reasonably be produced in available materials operating at moderate magnetic bias field levels.

III. Z_0 RATIO AND REFLECTION-COEFFICIENT DEPENDENCE ON FARADAY ROTATION LEVEL

The amount of Faraday rotation $\Delta\theta$ in a given length l of ferrite is equal to half the insertion phase difference between the RHCP and LHCP normal modes, i.e.,

$$\Delta\theta = \frac{\beta_R l - \beta_L l}{2}. \quad (8)$$

The relative activity of the Faraday rotation medium can be expressed as the amount of rotation per unit insertion phase. Since the insertion phase θ_0 for a linearly polarized wave is the average of the RHCP and LHCP insertion phases, i.e.,

$$\theta_0 = \frac{\beta_R l + \beta_L l}{2} \quad (9)$$

it follows that the normalized rotation will be

$$\frac{\Delta\theta}{\theta_0} = \frac{\beta_R - \beta_L}{\beta_R + \beta_L}. \quad (10)$$

Recall that since $\beta = \omega\sqrt{\mu\epsilon}$ and $Z_0 = \sqrt{\mu/\epsilon}$, changes in the wave impedances and propagation factors are directly proportional for changes only in the microwave magnetic permeability so that using (4),

$$\frac{\Delta\theta}{\theta_0} = \frac{Z_{0R} - Z_{0L}}{Z_{0R} + Z_{0L}} = \frac{\alpha^2 - 1}{\alpha^2 + 1}. \quad (11)$$

Next, solve for α^2 and recall that α^2 is the desired ratio of wave impedances, expressed in terms of the normalized Faraday rotation

$$\frac{Z_{0R}}{Z_{0L}} = \alpha^2 = \frac{1 + \frac{\Delta\theta}{\theta_0}}{1 - \frac{\Delta\theta}{\theta_0}}. \quad (12)$$

Finally, this result can be substituted into (7) to get the desired expression of optimum reflection as a function of normalized Faraday rotation. Fig. 3 shows a plot of the dependence of return loss of the “anomalous” orthogonal reflected wave on the normalized Faraday rotation when the co-polarized reflection is cancelled, i.e., when $r = 1$. Although the plot has been shown for normalized rotation values up to 0.5, the maximum values for practical devices are likely to be 0.1 or less. Consequently, the return loss for the orthogonal wave should be at least 26 dB below the amplitude of the linearly polarized incident wave.

IV. EXPERIMENTAL RESULTS

By extension from previous studies [4], [5] and from practical hardware designs, the analysis and discussion above may also be applied to finite structures realized in square or circular waveguides. To verify the computed behavior, a test structure operating at X-band was built, consisting of a long cylindrical ferrite rod of 0.280-in (7.11 mm) diameter metallized by sputtering, with each end matched to a square waveguide by a dielectric impedance transformer. A solenoidal coil of 3000 turns

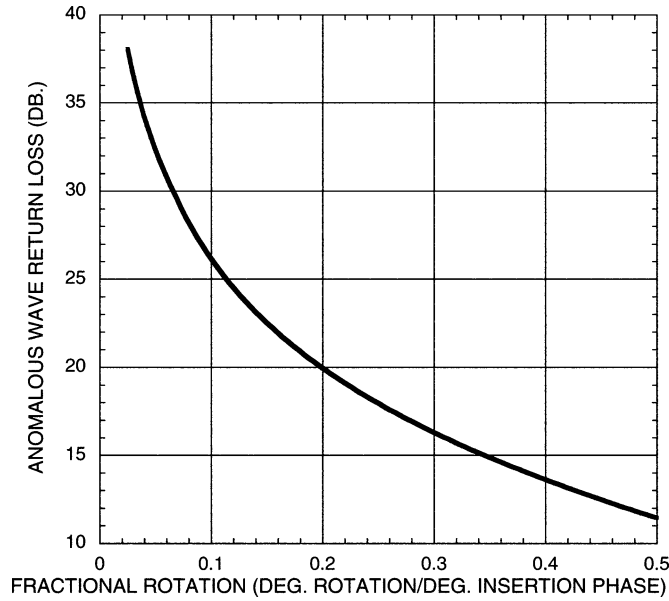


Fig. 3. Reflected wave return loss versus Faraday rotation level.

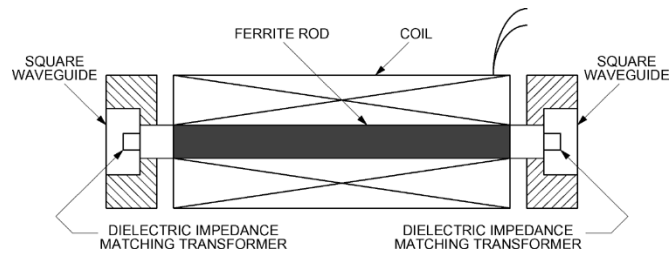


Fig. 4. Test structure configuration.

surrounded the ferrite rod, allowing the longitudinal magnetic bias field to be adjusted by varying the coil current. Fig. 4 shows a sketch of the configuration.

A test signal was applied to the ferrite rod through a rectangular waveguide port of a conventional orthogonal mode transducer (OMT) with cross-polarized reflected signals available at the second rectangular waveguide port of the OMT. Since there will be a cross-polarized reflection at the output of the rod, as well as the input, it was decided to use a ferrite with high loss at the test frequency in order to suppress the output reflected wave signal. An Ni ferrite with high saturation magnetization, Trans-Tech type TT 2-111 was selected. This material has a value of $\gamma 4\pi M_s$ equal to approximately 14 GHz, ensuring that the material would be well into the low-field loss region at 9 GHz. In fact, the attenuation in the rod of 3.4 in (86.4-mm length) was so great that there was no detectable signal at the output port when a normal measurement-level excitation was applied to the input. A photograph of the test hardware and setup is presented in Fig. 5.

Measured data from this fixture are presented in Fig. 6. It is clear that an orthogonal wave is reflected from the apparatus, and that the amplitude of this wave increases as the applied magnetic bias field is increased. Note that the onset of saturation effects in the ferrite material is indicated by the bend in the isolation versus current characteristic at a current level of approximately 140 mA. The Ni ferrite used saturates slowly, beginning

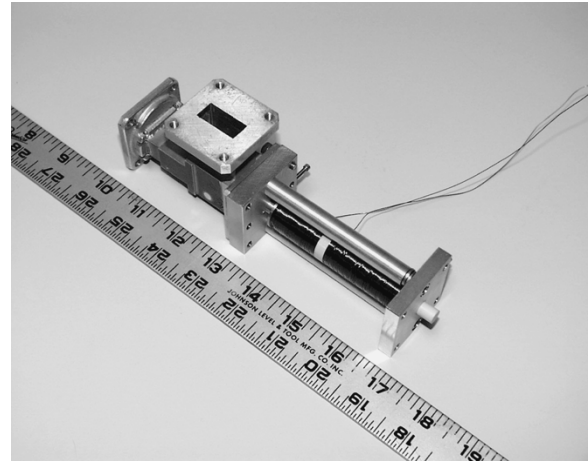


Fig. 5. Test hardware and setup.

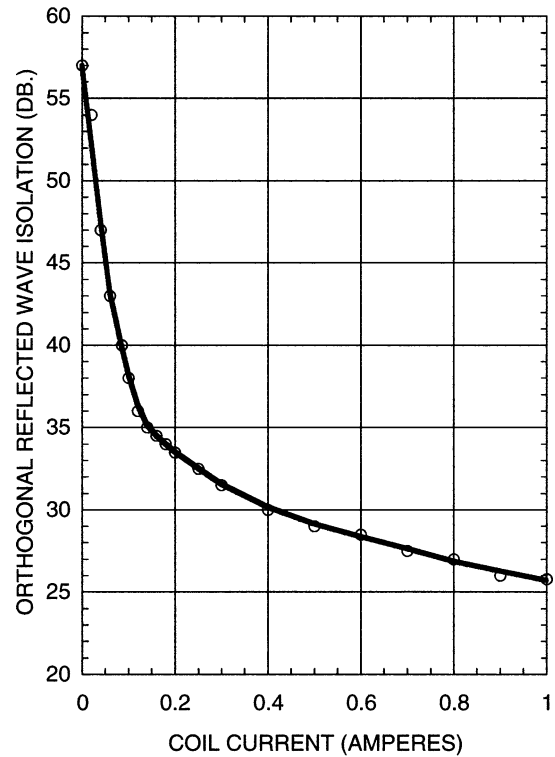


Fig. 6. Measured reflected wave return loss versus coil current.

at a magnetization level only approximately 0.5 times the saturation magnetization value. The value of isolation at zero current represents the intrinsic isolation of the OMT, which was very high in the vicinity of the test frequency.

V. CONCLUSIONS

The data presented in Figs. 3 and 6 allow the following conclusions to be made regarding the interface between fully filled square or round waveguides with and without Faraday rotation.

- 1) Impedance matching should proceed in the usual manner, minimizing the co-polarized reflection of the wave incident from the ordinary guide to the guide with Faraday rotation.

- 2) A small reflected wave will appear cross-polarized to the incident wave. Provisions should be made to absorb this wave in a film load or by coupling it to the side arm of an OMT.
- 3) The insertion loss increase caused by this orthogonal reflected wave should be limited to a few hundredths of a decibel.

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